

Diffusion Biased by a Soft Neck Linker Regulates Kinesin Stepping

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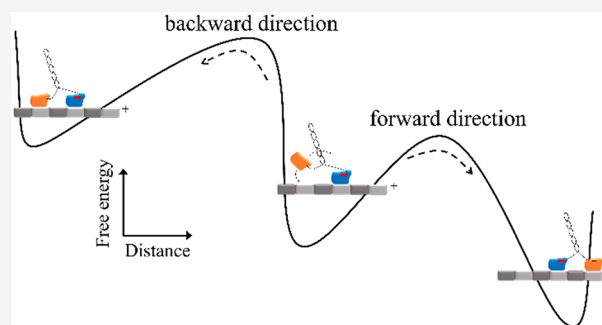
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ABSTRACT: Conventional kinesin is a high-performance motor that moves primarily toward the plus end of microtubules and occasionally toward the opposite direction. The physical mechanism of this directional stepping remains unclear. Here we develop a kinetic two-cycle model incorporating kinesin forward and backward stepping, in which the neck linker zippering and ATP catalysis process are conserved in backward steps. This model is quantitatively validated by a variety of experimental data, including load dependence of velocity, stepping ratio, and dwell time. The physical mechanism of kinesin stepping regulated by a biased diffusion process is identified by analyzing the load dependence and relevant thermodynamic properties of the model. Furthermore, the model suggests the kinesin directionality is optimized resulting from fulfilling a thermodynamic constraint. Our modeling provides a chemomechanical coupling mechanism that connects the flexibility of the neck linker zippering effect for direction rectification and the measured performance into a consistent frame.



1. INTRODUCTION

Conventional kinesin (hereafter called “kinesin”) is a high-performance molecular motor which uses the chemical energy released from ATP hydrolysis to generate directional movement along a polar track of microtubules (MTs).^{1,2} The motor comprises a pair of identical heads connected by two flexible neck linkers (each consists of ~ 14 amino acids).^{3,4} With this structure, kinesin moves toward the plus end of MTs in a hand-over-hand manner^{5–8} and takes one step of ~ 8 nm distance by hydrolyzing one ATP molecule.^{9–12} Single-molecular studies show that the directionality of kinesin is robust. Kinesin successively produces ~ 100 forward steps per second and rarely shows backward steps.^{10–12} Resisting loads increase in the probability of backward stepping and completely stall the motor at a force of ~ 7 pN.^{10–13} Moreover, a single-molecule experiment has detected processive backward movement under superstall loads (~ 15 pN).¹² A well-known direction-rectification mechanism of kinesin is called “neck linker zippering”,^{3,4,14,15} which produces a forward (plus-end) bias for the motor and thereby directs the detached head to move forward. However, this zippering configuration change only has a free energy gain of $\sim 1.2k_B T$,⁴ which is insufficient to support motor’s robust directionality. Though a number of theoretical models have been proposed,^{14–26} the physical mechanism of directional stepping remains controversial.

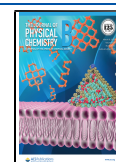
Experimental measurements showed that kinesin backward steps also required ATP binding.^{11,12} Moreover, the measured $\sim 70\%$ energy efficiency indicated that the backward step is not coupled with ATP synthesis induced by external loads, i.e., a

simple reverse of the forward step pathway. These discoveries suggested that the forward and backward steps may be connected by the ATP-driven process, which might be an interesting entry for understanding the physical mechanism of kinesin stepping. A kinetic model first proposed that experimental data on kinesin performance can be explained by introducing a second pathway for backward steps that was coupled with ATP hydrolysis.²⁰ However, the inspiration of this kinetic model is based on network analysis, which did not account for the significance of neck linker zippering for kinesin stepping. The neck linker is a very soft peptide chain with a persistence length of less than 1 nm, whose mechanical capacity for a forward step is likely limited. A recent study^{27,28} based on an exact solution of the wormlike chain model suggested that the forward bias induced by the neck linker zippering is approximately proportional to $\sim 7k_B T$. The resisting force can reduce or even reverse the forward bias to regulate kinesin directional stepping. In addition to the effect of neck linker zippering, the remarkable performance of kinesin depends on the working mechanisms of selective catalytic sites and strain-based gating.^{13,29} Recently, a research group proposed a series of improved models that integrated the

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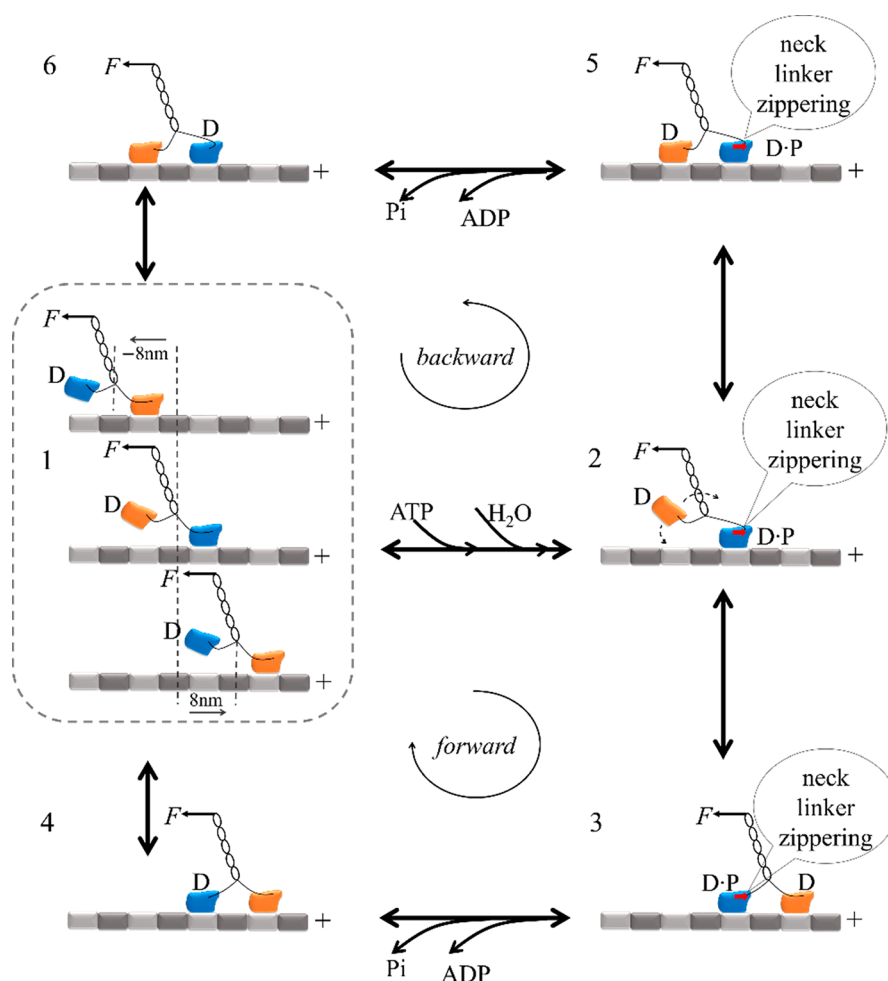


Figure 1. Modeling cycle for forward and backward stepping of kinesin. Kinesin's two heads are colored orange and blue. Microtubules are shown by a periodic repeated array of gray boxes, and the symbol "+" represents the plus end of the track. The two thin lines connecting the two heads are neck linkers, and the red arrow in a head represents the neck linker full zippering. F is the applied force, and the black arrow denotes the plus direction of applied force. "D" and "D-P" are the short forms of ADP and ADP-Pi. The motor's mechanical configuration and nucleotide bound states corresponding to each state are described in section 2. The three states in the gray dashed box are the same state but taking an -8 or 8 nm step.

mechanisms mentioned above, providing consistent and quantitative explanations of the available diverse experimental data.^{30–33} Thus, the neck linker zippering and ATP catalysis activity are possibly conserved in backward steps as they are in forward steps. Such a picture of an integrated mechanical and kinetic mechanism can be formed and may have the capacity of addressing kinesin directional stepping.

On the basis of above discussion, we propose a two-cycle kinetic model, in which the neck linker zippering and ATP catalysis process are conserved in backward steps. The model qualitatively reproduces a variety of experimental data collected from different experiments on kinesin.^{11,12,24,34} The model also qualitatively explains the performance reduction of kinesin caused by extending the neck linker.^{24,34} Our modeling suggests the directional stepping of kinesin is mainly regulated by a biased diffusion process. In addition, we find in our model the directionality of kinesin stepping is maximized due to fulfilling a thermodynamic constraint proposed by an earlier study.²³

2. THE MODEL AND CONSTRAINTS OF TRANSITION RATES

2.1. The Model. Our modeling of both forward and backward stepping cycles (Figure 1) starts with an ATP-waiting state (state 1),²⁴ in which the empty front head attached to MTs awaits ATP binding, while the other ADP-bound head remains detached from MTs. The process of ATP binding and hydrolysis leads to a conformational change of zippering the neck linker (symbolized as red arrows in states 2, 3, and 5) toward the plus end of MTs ($1 \rightarrow 2$). The zippering effect generates a forward diffusion bias that directs the detached head to bind forward ($2 \rightarrow 3$). However, the backward binding ($2 \rightarrow 5$) is still possible but with less probability due to the forward diffusion bias. The resisting force toward the minus end of MTs can reduce the forward diffusion bias and change the relative probabilities for the forward step and backward step. The two transition branches ($2 \rightarrow 3$ and $2 \rightarrow 5$) from state 2 form two stepping pathways ($1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$ and $1 \rightarrow 2 \rightarrow 5 \rightarrow 6 \rightarrow 1$) for forward and backward steps, respectively.

In the forward stepping pathway, the neck linker zippering directs the detached head to bind a forward site (Figure 1, state

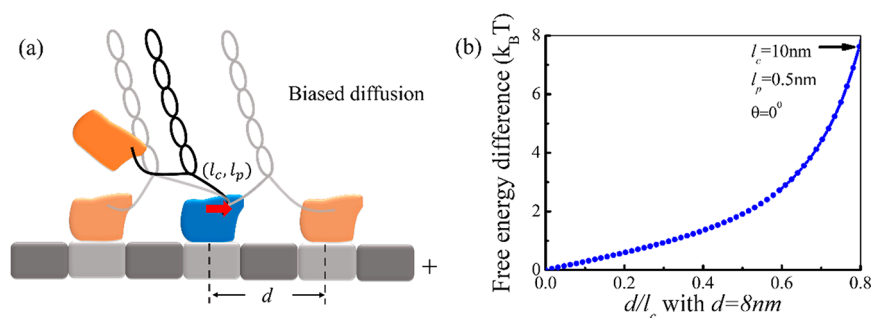


Figure 2. (a) Mechanical polymer representation of diffusion transition process induced by an effect of neck linker zippering (red arrow). The gray shadows represent two possible states corresponding to the diffusion head binding a forward or backward site, respectively. l_c and l_p correspond to the neck linker contour length and persistence length. (b) Diffusion bias exactly calculated based on a wormlike chain model with $l_c = 10$ nm, $l_p = 0.5$ nm, and $\theta = 0^\circ$; see details in ref 27.

3). Adjacently, the tethered head takes the leading position and releases ADP and then the rear head releases inorganic phosphate (Pi) (state 4). The rear head detaches from MTs, returning the motor to initial state, but taking an 8 nm forward step by hydrolyzing one ATP molecule. However, the detached head may rebind to the original site under high loads (state 5). Next, the rear head releases ADP, and sequentially Pi releases from the front head (state 6). The front head detaches from MTs and then binds to the backward site under high loads, completing an 8 nm backward step and hydrolyzing one ATP molecule as well. In addition to backward and forward pathways induced by ATP hydrolysis, there is a pathway for directional movement driven by applied force, i.e., $1 \rightarrow 2 \rightarrow 5 \rightarrow 6 \rightarrow 1$. Along the clockwise direction of this pathway, forward loads will induce forward steps. Conversely, backward steps will occur under backward loads following the anticlockwise direction.

2.2. Constraints of Transition Rates. The model is connected by seven transitions, and the rates of each transition are reversible (k_{ij} and k_{ji}). The transition rates are constrained by the molar concentration and the load force F (setting resisting force as a positive vector), which obeys a Boltzmann relationship.³⁵ Thus, the general form of the transition rates k_{ij} from state i to state j can be written as

$$k_{ij} = k_{ij}^0 [C]_{ij} \exp(-\delta_{ij} \bar{F}) \quad (1)$$

Here, k_{ij}^0 is the rate constant under equilibrium. $[C]_{ij}$ denotes the molar concentration with $C = \text{ATP}$, ADP , and Pi , and $[C]_{ij} = 1$ if the transition $i \rightarrow j$ involves no nucleotide binding. δ_{ij} and $\bar{F} = Fd/(k_B T)$ are a dimensionless characteristic distance and a dimensionless force, respectively. All the transition rates following this form are displayed in the Appendix. Furthermore, all transition rates obey the detailed-balance condition, which generates two constraints for the forward and backward stepping cycles, namely

$$k_B T \ln \left(\frac{k_{12} k_{23} k_{34} k_{41}}{k_{21} k_{32} k_{43} k_{14}} \right) = \Delta\mu - Fd$$

$$k_B T \ln \left(\frac{k_{12} k_{25} k_{56} k_{61}}{k_{21} k_{52} k_{65} k_{16}} \right) = \Delta\mu + Fd \quad (2)$$

Here, k_B is the Boltzmann constant and T is room temperature. $\Delta\mu = k_B T \ln \{K_{\text{eq}}[\text{ATP}]/([\text{ADP}][\text{Pi}])\}$ is the chemical energy released from hydrolyzing one ATP molecule,

where $K_{\text{eq}} = 4.9 \times 10^{11} \mu\text{M}$ is the equilibrium constant. d is the step distance of 8 nm.

On the basis of fitting results, the kinesin steps occur in a sequence of load-independent chemical transitions and load-dependent mechanical transitions. The rates of chemical transitions referring to nucleotide binding and release were detected by experiments.^{36,37} It should be noted that the rate k_{12} is mainly limited by ATP binding due to fast ATP hydrolysis. The mechanical transition branches $2 \rightarrow 3$ and $2 \rightarrow 5$ correspond to the diffusion process of a tethered head, which determines the stepping direction (Figure 2a). A recent theory based on a wormlike chain model indicated that the diffusion of the detached head can be biased by the neck linker zippering.²⁷ The diffusion bias for displacing the detached head to the forward site over the backward site can be quantified by an energy difference. We calculate the energy difference as a function of the contour length of the neck linker under zippering orientation parallel to microtubules ($\theta = 0^\circ$) as shown in Figure 2b. The free energy difference is approximately equal to $\sim 7k_B T$ under the parameters relevant to the neck linker of wild type kinesin (K-WT), $l_c = 10$ nm and $l_p = 0.5$ nm. For the mutant kinesin with extending neck linker (inserted 6 amino acid, called K-AA), the longer l_c leads to the reduction of the free energy difference consistent with experiments. In this study, we define the ratio of k_{23}/k_{25} as the diffusion bias γ , which is the load dependence obeying the equation

$$\gamma = \frac{k_{23}}{k_{25}} = \frac{k_{23}^0 \exp(-\delta_{23} \bar{F})}{k_{25}^0 \exp(\delta_{25} \bar{F})}$$

$$= \exp \left(\frac{G_0}{k_B T} \right) \exp[-(\delta_{23} + \delta_{25}) \bar{F}] \quad (3)$$

The diffusion bias is exponentially proportional to a free energy G_0 with no loads. Also, other mechanical transitions ($4 \rightarrow 1$ and $6 \rightarrow 1$) denote the head dissociation processes, which are load-independent suggested by fitting results. Thus, the characteristic distances should satisfy $\delta_{23} - \delta_{32} = 1$ and $\delta_{52} - \delta_{25} = 1$ according to eq 2. The fitting details are shown in the Appendix.

3. RESULTS AND DISCUSSION

3.1. Analytical Expressions of Kinesin Performance.

Within the framework of the kinetic–mechanical model, the analytical expressions of kinesin mean velocity, stepping ratio

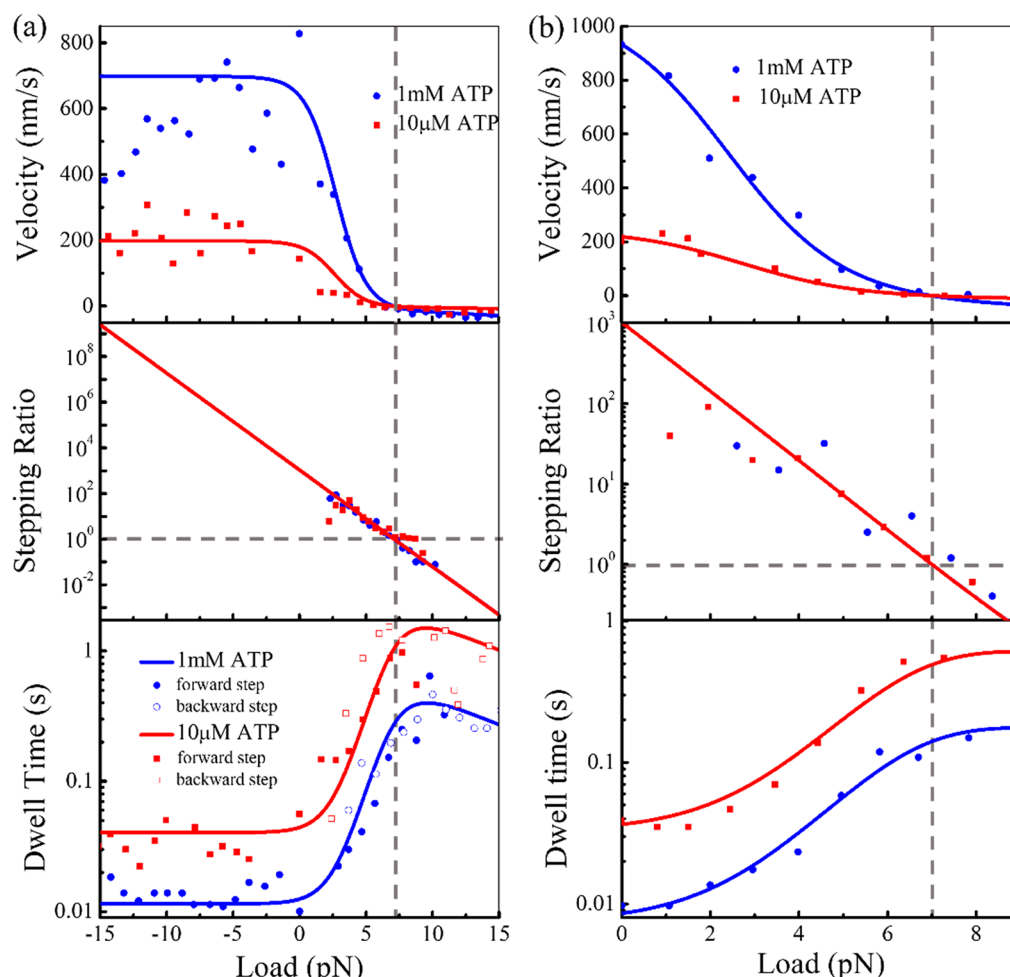


Figure 3. Theory–experiment comparison for K-WT. In the legends, blue and red symbols correspond to experimental ATP concentrations of 1 mM and 10 μ M, respectively. (a) Comparison with data from ref 12. (b) Comparison with data from ref 11. The stepping ratios in (a) and (b) for different ATP concentrations overlap each other. The vertical dashed lines correspond to the stall force. The fitting parameters are listed in Table 1.

(the number of forward steps relative to that of backward steps), and dwell time are derived.

3.1.1. Velocity. At steady state, the motor mean velocity can be written in terms of transition fluxes $v = d(J_f - J_b)$, where $J_f = p_2k_{23} + p_5k_{52}$ and $J_b = p_3k_{32} + p_2k_{25}$ denote respectively the number of forward and backward steps per unit time and p_i is the probability of the motor occupying state i . The velocity can be further derived in terms of entropy productions of $\Delta S_{ij} = k_B \ln(p_i k_{ij}/p_j k_{ji})$ and the diffusion bias γ for convenient thermodynamic analysis, which is given by

$$\frac{v}{d} = p_2 k_{23} \left\{ 1 - \exp\left(\frac{-\Delta S_{23}}{k_B}\right) + \gamma^{-1} \left[\exp\left(\frac{-\Delta S_{25}}{k_B}\right) - 1 \right] \right\} \quad (4)$$

3.1.2. Stepping Ratio. The stepping ratio (denoted by R) is a key feature to identify kinesin stepping, quantitatively given by $R = J_f/J_b = (p_2k_{23} + p_5k_{52})/(p_3k_{32} + p_2k_{25})$. Following the same rule as introduced into the velocity, R can be expressed as

$$R = \frac{1 + \gamma^{-1} \exp(-\Delta S_{25}/k_B)}{\gamma^{-1} + \exp(-\Delta S_{23}/k_B)} \quad (5)$$

3.1.3. Dwell Time. The average dwell time of each step under load force can be quantified as $\tau = (J_f - J_b)d/[(J_f + J_b)v]$. Here, it should be emphasized that this dwell time is average

dwell time for each step not referring to forward and backward steps explicitly. Based on the expressions of velocity and stepping ratio, τ can be rewritten as

$$\tau = \frac{(R - 1)d}{(R + 1)v} \quad (6)$$

Equations 4, 5, and 6 clearly show that motor performance strongly depends on the diffusion bias and entropy productions of mechanical transitions, which will be discussed in detail in section 3.4.

3.2. Model Reliability Examined by Theory–Experiment Comparison. To examine the reliability of our model, we first reproduce the experimental data from refs 11 and 12, which gives the performance of K-WT as a function of load force at different ATP concentrations of 1 mM and 10 μ M. The theory–experiment comparison shows that our theory quantitatively fit all the experimental data (Figure 3). The values of the parameters for fitting are displayed in Table 1. Notably, the characteristic distance δ_{ij} of each transition rate uncovers the central mechanism of kinesin load-dependence stepping movement. The fitting results reveal that the chemical transition should be independent or lightly independent of load force as the characteristic distances δ_{ij} for chemical transition rates are needed to be nearly equal to 0.

Table 1. Fitting Parameters^a

parameter	ref 12	ref 11	ref 22	ref 30, WT	ref 30, 6AA
k_{12}^0 ($\mu\text{M s}^{-1}$)	3.5	4	2	2	2
k_{34}^0 and k_{41}^0 (s^{-1})	180	280	100	195	160
k_{23}^0 (s^{-1})	1000	900	400	4500	2000
G_0 ($k_B T$)	7	7	<i>a</i>	7	6.5
δ_{23}	0.45	0.4	0.5	0.5	0.5

^a $k_{21}^0 = 100 \text{ s}^{-1}$, $k_{43}^0 = k_{56}^0 = 2 \times 10^{-4} \mu\text{M s}^{-1}$, $k_{14}^0 = k_{16}^0 = 2 \text{ s}^{-1}$, and $\delta_{12} = \delta_{21} = \delta_{34} = \delta_{43} = \delta_{56} = \delta_{65} = 0$ are all the same for different experiment groups. G_0 is equal to $2.2k_B T$, $3.4k_B T$, and $4.2k_B T$ under ATP concentrations of $2 \mu\text{M}$, $10 \mu\text{M}$, and 2 mM in the setting of ref 24. The rate constants for nucleotide binding and release are collected from experimental probing.^{36,37} These values are applied to the experiment data from refs 11, 12, 24, and 34, and the fitting results are displayed in Figures 3, 4, and 6.

Next, we discuss the diffusion process of the tethered head in an ATP-waiting state. In the case of no loads, the diffusion process is constrained by the effect of neck linker zippering (Figure 2a). A predication suggested by a mechanical study,²⁷ the neck linker zippering has the capability for transducing a diffusion bias exponentially proportional to $\sim 7k_B T$ (see Figure 2b). Our fitting is consistent with this predication. When a resisting load is imposed on the motor, the diffusion bias can be reduced or even reversed (eq 3), which has been verified by the mechanical study (for details, see Figure 4 in ref 27). According to eq 3, the load dependence of the diffusion bias is determined by the term of $\delta_{23} + \delta_{25} = 0.5$ with δ_{25} being a small positive number (0.05 and 0.1 for fitting the data of refs 12 and 11; see Table 1). Thus, the load force acts mainly on the forward binding rate k_{23} rather than on the backward binding rate k_{25} , which result is consistent with previous predictions.^{11,14} To further test this feature, we compare the theory predictions with experimental data collected from K-6AA with an extending contour length (Figure 4).^{24,34} The diffusion bias of K-6AA is smaller than that of K-WT. The fitting results bring a load-independence rate k_{25} , i.e., $\delta_{25} = 0$ and $\delta_{23} = 0.5$.

In addition, the mechanical transitions of $4 \leftrightarrow 1$ and $6 \leftrightarrow 1$ are independent of load for either K-WT or K-6AA. Thus, the load dependence of kinesin stepping is mainly regulated by the biased diffusion process. In accordance with Arrhenius kinetics, the transition rates depend on the relative activation energy of the barrier potential. Thus, the physical analysis of the kinesin directional stepping under both resisting and assisting load force regimes can be achieved by means of a free-energy

landscape picture (Figure 5). In Figure 5, the black solid line corresponds to the potential profile when no force is attached.

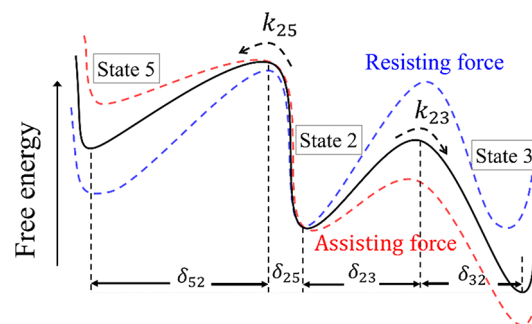


Figure 5. Schematic of free energy barriers to be overcome by the diffusion head binding backward or forward. The lines colored black, blue, and red represent the cases of energy barriers under no force, resisting force, and assisting force. See the discussion in the text.

On the basis of the characteristic distances δ_{23} and δ_{25} derived from the fitting (see Table 1), the resisting load force (0–7 pN) raises the forward barrier significantly, but has little influence on the backward barrier, displayed by the blue line in Figure 5. Until force goes to the stall force of ~ 7 pN, the backward barrier is the same as the forward potential, resulting in the loss of directionality of kinesin movement. In the regime of the superstall force (7–15 pN), the forward barrier continuously goes up, which will raise backward stepping. By contrast, in the regime of assisting load force (–15 to 0 pN), the forward barrier goes down, whereas the backward barrier has no significant change, which corresponds to the red line in Figure 5. Meanwhile, this energy profile leads to an increase in the diffusion bias with loads, as well as the stepping ratio (Figure 3a, middle panel). Although the stepping ratio increases, the motor velocity is saturated due to the limitation from other transition rates. For specificity, the limitations primarily come from the binding of ATP and the release of ADP and Pi for the cases of 1 mM ATP concentration and 10 μM ATP concentration, respectively.

Though K-6AA and K-WT share the same load dependence of the stepping mechanism, there is one major difference. The experiments on K-6AA found that a decrease in ATP concentration will effectively reduce the stall force but not change the load dependence of the stepping ratio (Figure 4b), i.e., ATP concentration dependence of stall force. Thus, the

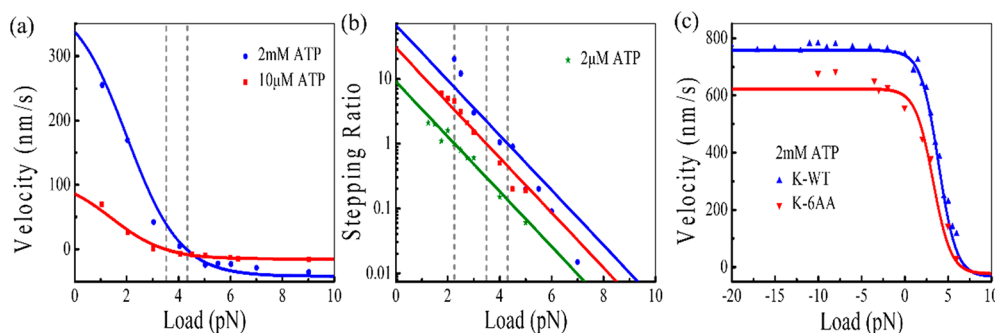


Figure 4. Theory–experiment comparison for K-6AA. (a) Velocity and (b) stepping ratio, compared with the data from ref 24 at different ATP concentrations of 2 mM, 10 μM , and 2 μM . (c) Velocity compared with data from ref 34 at saturated 2 mM ATP concentration. The blue and red curves correspond to K-WT and K-6AA, respectively. The vertical dashed lines correspond to the stall force. The fitting parameters are listed in Table 1.

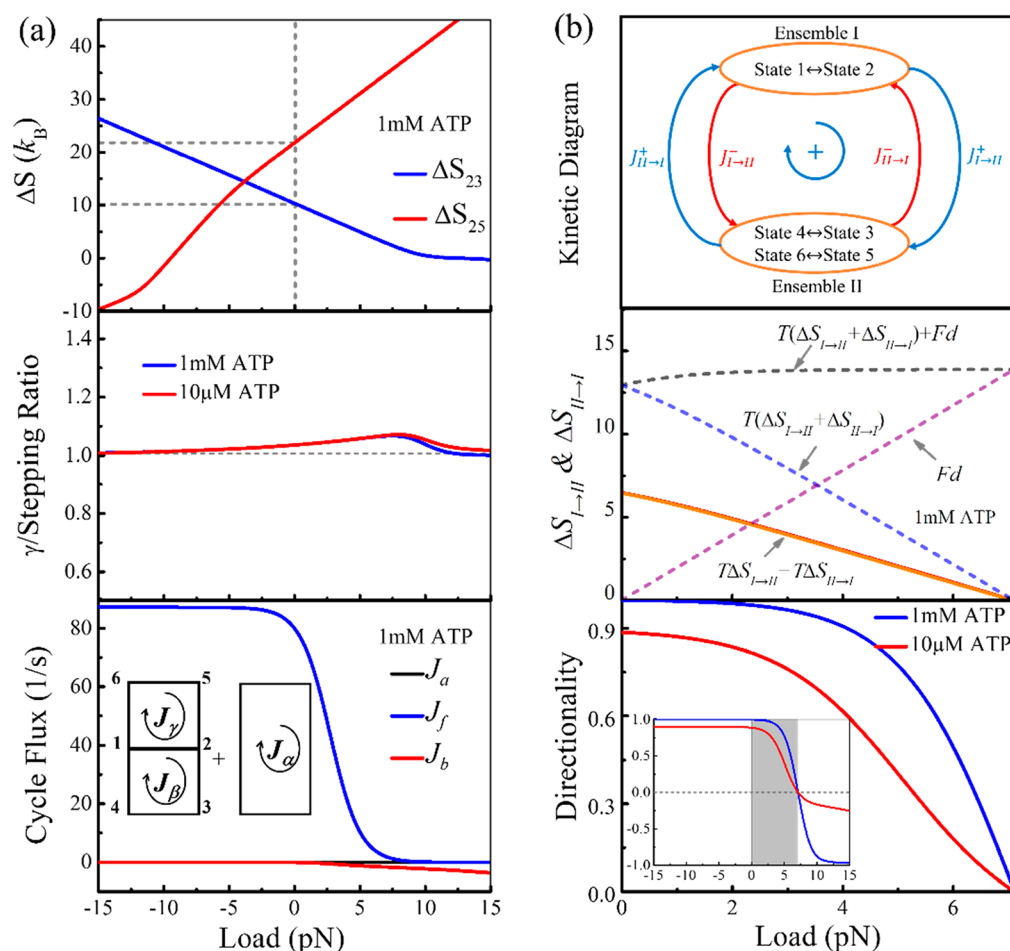


Figure 6. (a) Upper panel: Entropy productions of two mechanical transition pathways as a function of load force at 1 mM ATP concentration. Middle panel: The ratio between the diffusion bias and stepping ratio as a function of force at different ATP concentrations. Lower panel: The three cycle fluxes as displayed in the inset vary with load force. (b) Upper panel: The kinetic diagram by grouping the mechanical transition fluxes into four types. Middle panel: Two weighted average entropy productions and mechanical work output as a function of force. Lower panel: The expansion for the gray shadow section of the inset that was directionality calculated by eq 7 as a function of force at different ATP concentrations. Above results are derived based on ref 12; see the main text.

value of G_0 decreases with ATP concentration derived from eq 3. For instance, the values of G_0 for fitting in Figure 4a,b are equal to $2.2k_B T$, $3.4k_B T$, and $4.2k_B T$ at ATP concentrations of 2 μ M, 10 μ M, and 2 mM, respectively (see Table 1). Extending the contour length gives rise to the reduction of interhead tension in the two-head bound state, which leads to a smaller G_0 compared with K-WT. This feature is identified by the mechanics of our model (Figure 2), which predicts that the unloaded diffusion bias γ will reduce with increase in the contour length l_c of neck linker.

A qualitative understanding of the ATP concentration dependence of the stall force can be achieved. According to eq 5, the stepping ratio also depends on the average energy supply associated with ATP concentration (see $\Delta\mu$ in eq 2). To put it clearly, the neck linker zippering requires energy input, but the extended neck linker decreases in the efficiency of energy conversion.²⁷ However, the quantitative explanation of how the stepping ratio depends on ATP concentration requires further experimental detection. For instance, the experimental measurements should provide clear information about the energy supply with control of the concentrations of ADP and Pi.

3.3. Thermodynamic Analysis of the Model. In this section, we systematically explain the load dependence of kinesin directional stepping by means of thermodynamic analysis under the fitting parameters of ref 12. In accordance with eqs 4–6, the load dependence of the motor's performance depends on the two entropy productions (ΔS_{23} and ΔS_{25}) as well as the diffusion bias γ . The upper panel of Figure 6a plots the two entropy productions as a function of load force at 1 mM ATP concentration. In the assisting regime ($-15 \text{ pN} < F < 0 \text{ pN}$), the term of $\exp(-\Delta S_{23}/k_B)$ in eqs 4 and 5 can be omitted because the entropy production ΔS_{23} is always larger than $10k_B T$ (see Figure 6a). Besides, the term γ^{-1} approaches 0 in the assisting regime according to eq 3. Thus, eqs 4 and 5 can be approximately simplified into the forms of $v/d \approx p_2 k_{23}$ and $R \approx \gamma$, respectively. In contrast to the case of resisting regimes ($0 \text{ pN} < F < 15 \text{ pN}$), the entropy production ΔS_{25} increases as the load force goes up to 15 pN, which implies that the value of $\exp(-\Delta S_{25}/k_B)$ goes to 0. Meanwhile, the entropy production ΔS_{23} continually decreases to 0, giving rise to $\exp(-\Delta S_{23}/k_B) = 1$ when the loads surpass the stall force. The term γ^{-1} is larger than 1 when $F > 7 \text{ pN}$. In the regime $F > 7 \text{ pN}$, eqs 4 and 5 can be simplified into $v/d \approx -p_2 k_{25}$ (the minus sign means the motor proceeds toward the minus end of MTs)

and $R \approx \gamma$, respectively. In order to further examine the approximation relation $R \approx \gamma$, we plot the ratio of γ/R at each force, as shown in the middle panel of Figure 6a. The plot reveals that the stepping ratio R always approximates γ regardless of load force, which gives the uniform exponential dependence of R on load force. However, the velocity is mainly determined by $v/d \approx p_2 k_{23}$ for small resisting load and assisting load, and by $v/d \approx -p_2 k_{25}$ for loads larger than the stall force. Hence velocity displays a more complicate and different load dependence than the stepping ratio.

In addition, the model consists of three cycle fluxes, namely J_α , J_β , and J_γ , as displayed in the inset of Figure 6a, lower panel. The motor directional stepping can be well described by the cycle fluxes.^{38,39} J_α denotes the cycle flux driven by applied force rather than ATP hydrolysis and is approximately equal to 0 under each force, which feature suggests that it is difficult to drive quick and continuous steps of kinesin by purely external force. The cycle flux J_β for forward stepping driven by ATP hydrolysis does not change very much under assisting force, but drops dramatically to 0 when resisting force goes beyond the stall force. The third cycle flux, J_γ , of backward stepping driven by ATP hydrolysis does not work until the force exceeds the stall force, which result is consistent with experimental detection.

3.4. Optimized Thermodynamic Constraint for Directionality. A kinetic–thermodynamic theory developed a universal method to analyze a motor’s directionality measuring the fidelity of the motor’s self-propelled motion toward the plus end direction.²³ Following this kinetic–thermodynamic treatment, we reform the kinetic model (Figure 1) into a kinetic diagram (the upper panel of Figure 6b), which is conveniently used to analyze the motor’s directionality. In this kinetic diagram, six states are categorized into two ensembles: One comprises all single-binding states (marked as ensemble I), in which only one head is MTs-bound; the other consists of all double-binding states (marked as ensemble II), with two heads bound to MTs. The motor’s directionality, defined by the equation of $D = P_+ - P_-$ with normalized probabilities for forward steps P_+ and for backward steps P_- , can be expressed in terms of transition fluxes. The mechanical transition fluxes in the kinetic diagram can be grouped into four types based on ref 23: $J_{I \rightarrow II}^+ = p_2 k_{23} + p_1 k_{16}$ and $J_{I \rightarrow II}^- = p_2 k_{25} + p_1 k_{14}$ denote the sum of mechanical transition fluxes for the detached head binding forward and backward sites from ensemble I to II, respectively; $J_{II \rightarrow I}^+ = p_4 k_{41} + p_5 k_{52}$ and $J_{II \rightarrow I}^- = p_3 k_{32} + p_6 k_{61}$ denote the sum of mechanical transition fluxes for the transitions of the rear or front head dissociation from MTs, respectively. The plus or minus sign in a superscript denotes a transition resulting in a position shift toward or away from the desired direction (plus end). Two weighted average entropy productions can be defined by the ratios of any two types of mechanical transition fluxes as follows: $\langle \exp(\Delta S_{I \rightarrow II}/k_B) \rangle = J_{I \rightarrow II}^+/J_{II \rightarrow I}^+$ and $\langle \exp(\Delta S_{II \rightarrow I}/k_B) \rangle = J_{II \rightarrow I}^-/J_{I \rightarrow II}^-$ (derivation details available from ref 23). Thus, the motor’s directionality can be expressed in terms of two weighted average entropy productions as follows (see the derivation in the Appendix):

$$D = \frac{(\langle \exp(\Delta S_{I \rightarrow II}/k_B) \rangle - 1)(\langle \exp(\Delta S_{II \rightarrow I}/k_B) \rangle - 1)}{\langle \exp(\Delta S_{I \rightarrow II}/k_B) \rangle \langle \exp(\Delta S_{II \rightarrow I}/k_B) \rangle - 1} \quad (7)$$

Using a one-way cycle for kinesin, the kinetic–thermodynamic theory derived a thermodynamic equality of two

weighted average entropy productions for optimizing the motor’s directionality.²³ Surprisingly, our theoretical calculation shows that the two weighted average entropy productions are approximately equal for each force and different ATP concentrations, i.e., $\Delta S_{I \rightarrow II} = \Delta S_{II \rightarrow I}$. Now, we focus on the resisting regime between 0 and 7 pN for analyzing kinesin’s optimized performance (the middle panel of Figure 6b). The most energy input of ATP hydrolysis is dissipated into heat to either maintain directionality movement or resist load force, i.e., mechanical work output, namely, $\Delta\mu_H = T(\Delta S_{I \rightarrow II} + \Delta S_{II \rightarrow I}) + Fd$. The middle panel of Figure 6b shows that the sum of two weighted average entropies decreases and the mechanical work output increases as the load goes to the stall force. However, the effective energy input $\Delta\mu_H$ (corresponding to the gray dashed line) is maintained at a constant level ($\sim 14k_B T$), which happens to correspond to the stall force of ~ 7 pN. The lower panel of Figure 6b plots the directionality as a function of force at ATP concentrations of 1 mM and 10 μ M. In the former (1 mM ATP) the directionality approaches 1 under the assisting regime and then sharply decreases to -1 in the superstall force range, which implies that the directionality toward the minus end is as robust as the directionality toward the plus end. However, in the case of 10 μ M ATP concentration, the directionality is smaller than -1 in the superstall force range, identifying a conclusion that backward steps need ATP binding.^{11,12}

4. CONCLUSION

As we know, the observed ATP dependence of backward stepping and measured $\sim 70\%$ energy efficiency indicate that the backward step is not coupled with ATP synthesis, i.e., a simple reverse of the forward step. Besides that, it is difficult to drive the process of backward stepping by purely external force unless the force is larger than 15 pN. However, the backward stepping with a rare proportion still exists even under the unloaded case. These clues may suggest the backward stepping also couples with ATP hydrolysis. It is natural that the neck linker zippering occurs in the backward stepping due to ATP binding and hydrolysis. Even in the presence of neck linker zippering, the soft neck linker cannot resist the load force as well as thermal fluctuation to rebind the backward site. Still owing to the flexibility of the neck linker, it is difficult for the load force to affect the zippering effect. Thus, the processes of ATP hydrolysis and neck linker zippering may be shared by forward and backward stepping. In this article, we introduce a two-cycle model incorporating kinesin forward and backward pathways. In this model, the backward stepping still conserves the activities of ATP hydrolysis and neck linker docking as they are in forward stepping.

The major findings of this study can be summarized as follows. First, load force primarily influences kinesin directional stepping via a biased diffusion process induced by neck linker zippering, which mechanism can be clearly described by a free-energy landscape picture. Second, our kinetic model fulfills a thermodynamic constraint that two weighted average entropy productions are equal at each force, which is an optimized sign for the motor’s directionality. Owing to the universality of the physical mechanism in the description of kinesin, it will be interesting to extend our approach in this study in the future to other soft polymer systems. From a broader perspective, the kinetic model we developed in this study has development potential evolving into a kinetic–polymeric model to

investigate more detailed information on the kinesin movement mechanism.

■ APPENDIX

A.1. Parametrization and Load Dependence of Transition Rates

In general, transition rates depend on molar concentrations of ATP, ADP, and Pi, as well as the load force F . According to eq 1 and the fitting process, the transition rates can be categorized as load-independent rates and load-dependent rates. The characteristic distance δ_{ij} of chemical transition rates should be in a narrow range of near 0 required by fitting. In addition, the mechanical transitions of $4 \leftrightarrow 1$ and $6 \leftrightarrow 1$ are also load independent or lightly load dependent. Thus, the first type of load-independent transition rates only relies on molar concentration and can thus be parametrized as in sequence:

$$\begin{aligned} k_{12} &= k_{12}^0[\text{ATP}], & k_{21} &= k_{21}^0 \\ k_{34} &= k_{34}^0, & k_{43} &= k_{43}^0[\text{ADP}][\text{Pi}] \\ k_{41} &= k_{41}^0, & k_{14} &= k_{14}^0 \\ k_{56} &= k_{56}^0, & k_{65} &= k_{65}^0[\text{ADP}][\text{Pi}] \\ k_{61} &= k_{61}^0, & k_{16} &= k_{16}^0 \end{aligned} \quad (\text{A1})$$

The values of most of the parameters in eq A1 are displayed in Table 1. The transition of $4 \rightarrow 3$ corresponds to the combination of Pi and ADP in turn. Hence, the transition time is determined by $t_{34} = 1/t_{34} = 1/k_{\text{ADPb}}[\text{ADP}] + 1/k_{\text{Pib}}[\text{Pi}]$, where k_{ADPb} and k_{Pib} are the constant rates of binding ADP and Pi, respectively. Since k_{ADPb} , k_{Pib} , $[\text{ADP}]$, and $[\text{Pi}]$ are fixed, k_{43} can be rewritten as $k_{43} = k_{43}^0[\text{ADP}][\text{Pi}]$ with $k_{43}^0 = (k_{\text{Pib}}^{-1}[\text{ADP}] + k_{\text{ADPb}}^{-1}[\text{Pi}])^{-1}$. The transition rate k_{56} is equal to k_{43} . Also, $k_{25}^0 = k_{23}^0 \exp(G_0/k_B T)^{-1}$ and k_{32}^0 and k_{52}^0 can be deduced from eq 2.

If eq 1 is inserted into eq 2, we can decompose eq 2 into two parts. Among that, the load-dependent part can be written as

$$\begin{aligned} k_B T \ln \left(\frac{\exp(-\delta_{23}\bar{F})}{\exp(-\delta_{32}\bar{F})} \right) &= -Fd \\ k_B T \ln \left(\frac{\exp(-\delta_{25}\bar{F})}{\exp(-\delta_{52}\bar{F})} \right) &= +Fd \end{aligned} \quad (\text{A2})$$

Based on eq A2, the relations of $\delta_{23} - \delta_{32} = 1$ and $\delta_{52} - \delta_{25} = 1$ can be deduced. Thus, all transition rates are settled and we can derive the analytical solutions of eqs 4–6.

A.2. Analytical Derivation for Directionality

As described in ref 23, in the kinetic model diagram of Figure 6b (upper panel), J_a^+ and J_a^- are the combined fluxes of all forward-step and backward-step cycles, and J_b (J_c) is the combined flux of futile-step cycles that correspond to rear (front) head dissociation followed by reverse binding. The four combined cycle fluxes can be expressed by the addition of related transition fluxes, i.e., $J_{1 \rightarrow 2}^+ = J_a^+ + J_b$, $J_{1 \rightarrow 2}^- = J_a^- + J_c$, $J_{2 \rightarrow 1}^+ = J_a^+ + J_c$, and $J_{2 \rightarrow 1}^- = J_a^- + J_b$. The directionality is $D = P_+ - P_-$, with P_+ and P_- as normalized possibilities for forward and backward steps. P_+ and P_- can be defined in terms of cycle fluxes as $P_+ = J_a^+/(J_a^+ + J_a^- + J_b + J_c)$ and $P_- = J_a^-/(J_a^+ + J_a^- + J_b + J_c)$. Therefore, the directionality can be rewritten in terms of the addition of transition fluxes, namely $D = (J_{1 \rightarrow 2}^+ - J_{2 \rightarrow 1}^-)/(J_{1 \rightarrow 2}^+ + J_{1 \rightarrow 2}^-)$. Owing to the steady-state approximation $J_{1 \rightarrow 2}^+ +$

$J_{1 \rightarrow 2}^- = J_{2 \rightarrow 1}^+ + J_{2 \rightarrow 1}^-$, D can be further expressed by two ratios of transition fluxes:

$$D = \frac{(J_{1 \rightarrow 2}^+/J_{2 \rightarrow 1}^- - 1)(J_{2 \rightarrow 1}^+/J_{1 \rightarrow 2}^- - 1)}{J_{1 \rightarrow 2}^+ J_{2 \rightarrow 1}^+ / (J_{2 \rightarrow 1}^- J_{1 \rightarrow 2}^-) - 1} \quad (\text{A3})$$

Using two weighted average entropy productions to substitute for the two ratios of transition fluxes, eq 7 can be derived.

A.3. Optimization Directionality Based on a Thermodynamic Constraint

As described in section 3.4, the fuel energy is converted into either mechanical work or energy dissipation to maintain directionality movement, namely, $\Delta\mu_H = T(\Delta S_{1 \rightarrow 2} + \Delta S_{2 \rightarrow 1}) + Fd$. Thus, the energy dissipation can be expressed as $\Delta\mu_D = T(\Delta S_{1 \rightarrow 2} + \Delta S_{2 \rightarrow 1}) = \Delta\mu_H - Fd$. Below, we will identify that the optimization directionality can be achieved when a thermodynamic constraint of $\langle \exp(\Delta S_{1 \rightarrow 2}/k_B) \rangle = \langle \exp(\Delta S_{2 \rightarrow 1}/k_B) \rangle = \exp(\Delta\mu_D/2k_B T)$ is satisfied.

For simplification, we use x , y , and C to substitute for $\langle \exp(\Delta S_{1 \rightarrow 2}/k_B) \rangle$, $\langle \exp(\Delta S_{2 \rightarrow 1}/k_B) \rangle$, and $\exp(\Delta\mu_D/2k_B T)$ in turn. By consideration of the relation of $xy = C$, eq 7 can be written as

$$D = \frac{(x - 1)(Cx^{-1} - 1)}{C - 1} \quad (\text{A4})$$

It is easy to acquire its derivations over x :

$$\frac{dD}{dx} = \frac{Cx^{-2} - 1}{C - 1} \quad (\text{A5})$$

When $x < C^{1/2}$, $dD/dx > 0$; when $x = C^{1/2}$, $dD/dx = 0$; when $x > C^{1/2}$, $dD/dx < 0$. Thus, $x = y = C^{1/2}$; the value of eq A3 is maximized. That is equal to saying that the equality of two weighted average entropy productions is an optimized sign for directionality.

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Notes

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■ REFERENCES

- (1) Vale, R. D.; Milligan, R. A. The way things move: looking under the hood of molecular motor proteins. *Science* **2000**, *288*, 88–95.
- (2) Vale, R. D. The molecular motor toolbox for intracellular transport. *Cell* **2003**, *112*, 467–480.
- (3) Rice, S.; Lin, A. W.; Safer, D.; Hart, C. L.; Naber, N.; Carragher, B. O.; Cain, S. M.; Pechatnikova, E.; Wilson-Kubalek, E. M.; Whittaker, M.; et al. A structural change in the kinesin motor protein that drives motility. *Nature* **1999**, *402*, 778–784.
- (4) Rice, S.; Cui, Y.; Sindelar, C.; Naber, N.; Matuska, M.; Vale, R. D.; Cooke, R. Thermodynamic properties of the kinesin neck-region docking to the catalytic core. *Biophys. J.* **2003**, *84*, 1844–1854.
- (5) Asbury, C. L.; Fehr, A. N.; Block, S. M. Kinesin moves by an asymmetric hand-over-hand mechanism. *Science* **2003**, *302*, 2130–2134.
- (6) Yildiz, A.; Tomishige, M.; Vale, R. D.; Selvin, P. R. Kinesin walks hand-over-hand. *Science* **2004**, *303*, 676–678.
- (7) Shao, Q.; Gao, Y. Q. On the hand-over-hand mechanism of kinesin. *Proc. Natl. Acad. Sci. U. S. A.* **2006**, *103*, 8072–8077.
- (8) Guydosh, N. R.; Block, S. M. Not so lame after all: kinesin still walks with a hobbled head. *J. Gen. Physiol.* **2007**, *130*, 441–444.
- (9) Schnitzer, M. J.; Block, S. M. Kinesin hydrolyses one ATP per 8-nm step. *Nature* **1997**, *388*, 386–390.
- (10) Visscher, K.; Schnitzer, M. J.; Block, S. M. Single kinesin molecules studied with a molecular force clamp. *Nature* **1999**, *400*, 184–189.
- (11) Nishiyama, M.; Higuchi, H.; Yanagida, T. Chemomechanical coupling of the forward and backward steps of single kinesin molecules. *Nat. Cell Biol.* **2002**, *4*, 790–797.
- (12) Carter, N. J.; Cross, R. A. Mechanics of the kinesin step. *Nature* **2005**, *435*, 308–312.
- (13) Yildiz, A.; Tomishige, M.; Gennerich, A.; Vale, R. D. Intramolecular strain coordinates kinesin stepping behavior along microtubules. *Cell* **2008**, *134*, 1030–1041.
- (14) Taniguchi, Y.; Nishiyama, M.; Ishii, Y.; Yanagida, T. Entropy rectifies the Brownian steps of kinesin. *Nat. Chem. Biol.* **2005**, *1*, 342–347.
- (15) Milic, B.; Andreasson, J. O.; Hancock, W. O.; Block, S. M. Kinesin processivity is gated by phosphate release. *Proc. Natl. Acad. Sci. U. S. A.* **2014**, *111*, 14136–14140.
- (16) Schnitzer, M. J.; Visscher, K.; Block, S. M. Force production by single kinesin motors. *Nat. Cell Biol.* **2000**, *2*, 718–723.
- (17) Fisher, M. E.; Kolomeisky, A. B. Simple mechanochemistry describes the dynamics of kinesin molecules. *Proc. Natl. Acad. Sci. U. S. A.* **2001**, *98*, 7748–7753.
- (18) Mogilner, A.; Fisher, A. J.; Baskin, R. J. Structural changes in the neck linker of kinesin explain the load dependence of the motor's mechanical cycle. *J. Theor. Biol.* **2001**, *211*, 143–157.
- (19) Tsygankov, D.; Fisher, M. E. Mechanoenzymes under superstall and large assisting loads reveal structural features. *Proc. Natl. Acad. Sci. U. S. A.* **2007**, *104*, 19321–19326.
- (20) Liepelt, S.; Lipowsky, R. Kinesin's network of chemomechanical motor cycles. *Phys. Rev. Lett.* **2007**, *98*, 258102.
- (21) Wang, Z.; Feng, M.; Zheng, W.; Fan, D. Kinesin is an evolutionarily fine-tuned molecular ratchet-and-pawl device of decisively locked direction. *Biophys. J.* **2007**, *93*, 3363–3372.
- (22) Hyeon, C.; Klumpp, S.; Onuchic, J. N. Kinesin's backsteps under mechanical load. *Phys. Chem. Chem. Phys.* **2009**, *11*, 4899–4910.
- (23) Efremov, A.; Wang, Z. Universal optimal working cycles of molecular motors. *Phys. Chem. Chem. Phys.* **2011**, *13*, 6223–6233.
- (24) Clancy, B. E.; Behnke-Parks, W. M.; Andreasson, J. O.; Rosenfeld, S. S.; Block, S. M. A universal pathway for kinesin stepping. *Nat. Struct. Mol. Biol.* **2011**, *18*, 1020–1027.
- (25) Ren, J. Detectable states, cycle fluxes, and motility scaling of molecular motor kinesin: An integrative kinetic graph theory analysis. *Front. Phys.* **2017**, *12*, 120505.
- (26) Xu, H. J.; Tong, T.; Hou, R. Z.; Li, H. R. Reconceptualizing kinesin's working cycle as separate chemical and mechanical processes. *Front. Phys.* **2018**, *13*, 138206.
- (27) Hou, R. Z.; Wang, N.; Bao, W.; Wang, Z. Mechanical transduction via a single soft polymer. *Phys. Rev. E: Stat. Phys., Plasmas, Fluids, Relat. Interdiscip. Top.* **2018**, *97*, 042504.
- (28) Hou, R. Z.; Wang, N.; Bao, W.; Wang, Z. Polymer-Based Accurate Positioning: An Exact Worm-like-Chain Study. *ACS Omega* **2018**, *3*, 14318–14326.
- (29) Rosenfeld, S. S.; Fordyce, P. M.; Jefferson, G. M.; King, P. H.; Block, S. M. Stepping and stretching - How kinesin uses internal strain to walk processively. *J. Biol. Chem.* **2003**, *278*, 18550–18556.
- (30) Guo, S. K.; Wang, P. Y.; Xie, P. A model of processive movement of dimeric kinesin. *J. Theor. Biol.* **2017**, *414*, 62–75.
- (31) Guo, S. K.; Wang, P. Y.; Xie, P. Dynamics of dimeric kinesins: Limping, effect of longitudinal force, effects of neck linker extension and mutation, and comparison between kinesin-1 and kinesin-2. *Int. J. Biol. Macromol.* **2017**, *105*, 1126–1137.
- (32) Guo, S. K.; Shi, X. X.; Wang, P. Y.; Xie, P. Force dependence of unbinding rate of kinesin motor during its processive movement on microtubule. *Biophys. Chem.* **2019**, *253*, 106216.
- (33) Xie, P. Theoretical Analysis of Dynamics of Kinesin Molecular Motors. *ACS Omega* **2020**, *5*, 5721–5730.
- (34) Andreasson, J. O.; Milic, B.; Chen, G.-Y.; Guydosh, N. R.; Hancock, W. O.; Block, S. M. Examining kinesin processivity within a general gating framework. *eLife* **2015**, *4*, e07403.
- (35) Walcott, S. The load dependence of rate constants. *J. Chem. Phys.* **2008**, *128*, 215101.
- (36) Cross, R. A. The kinetic mechanism of kinesin. *Trends Biochem. Sci.* **2004**, *29*, 301–309.
- (37) Valentine, M. T.; Gilbert, S. P. To step or not to step? How biochemistry and mechanics influence processivity in Kinesin and Eg5. *Curr. Opin. Cell Biol.* **2007**, *19*, 75–81.
- (38) Hill, T. L. Discrete-time random walks on diagrams (graphs) with cycles. *Proc. Natl. Acad. Sci. U. S. A.* **1988**, *85*, 5345–5349.
- (39) Hill, T. L. *Free Energy Transduction and Biochemical Cycle Kinetics*; Springer: New York, 1989.